Mushroom dataset

Load database

In [1]:
```
#STEP 1: LOAD DATABASE
import time
import pathlib
import Func_load_transaction_database as Func_load_DB
import psutil
import os
from datetime import datetime

now = datetime.now()
print("The last running =", now)

process = psutil.Process(os.getpid())

def memory_usage_mb():
    return process.memory_info().rss / 1024 / 1024  # MB

dbName = "Mushroom"  # Change this to the desired database name
start_time = time.time()
DBtrans = Func_load_DB.Func_load_database(dbName)
print("Time to load database from .dat: %s seconds ---" % (time.tim
print('Load Transaction database successfully')
# print(DBtrans)
print(f"RAM usage after loading DB: {memory_usage_mb():.2f} MB")

max_level = 0 # Maximum level of itemsets to mine

# Robust thresholds for TOI mining
Sup = 0
Occ = 0
Pro = 0
# decreasing thresholds for higher levels
# (these can be adjusted based on the dataset (densy/sparse)):
# List of thresholds for each level (level 1 to max_level)
volume_threshold = []
occupancy_threshold = []
revenue_threshold = []

# Sales-oriented strategy with parameters:
# C1 Uniform strategy: alpha = 1/3, beta = 1/3, gamma = 1/3
# C2 Support-oriented strategy: alpha = 1.0, beta = 0.0, gamma = 0.
# C3 Occupancy-oriented strategy: alpha = 0.0, beta = 1.0, gamma =
# C4 Profit-oriented strategy: alpha = 0.0, beta = 0.0, gamma = 1.0
# C5 Important support: alpha = 0.5, beta = 0.25 gamma = 0.25
# C6 Important occupancy: alpha = 0.25, beta = 0.5, gamma = 0.25
# C7 Important profit: alpha = 0.25, beta = 0.25, gamma = 0.5
# C8 Important support and occupancy: alpha = 0.5, beta = 0.5, gamm
# C9 Important support and profit: alpha = 0.5, beta = 0.0, gamma =
# C10 Important occupancy and profit: alpha = 0.0, beta = 0.5, gamm
```

```
sales_strategies = {
    "C1": (1/3, 1/3, 1/3),
    "C2": (1.0, 0.0, 0.0),
    "C3": (0.0, 1.0, 0.0),
    "C4": (0.0, 0.0, 1.0),
    "C5": (0.5, 0.25, 0.25),
    "C6": (0.25, 0.5, 0.25),
    "C7": (0.25, 0.25, 0.5),
    "C8": (0.5, 0.5, 0.0),
    "C9": (0.5, 0.0, 0.5),
    "C10": (0.0, 0.5, 0.5)
}

index_case = 9 # Change value to select different strategy (C1 to C

current_Case = sales_strategies["C"+str(index_case)]
# Set of bias factor
alpha = current_Case[0]
beta = current_Case[1]
gamma = current_Case[2]


flags = {
    "C1": (1, 1, 1),      # tri-objectives
    "C2": (1, 0, 0),      # single-objective: support-oriented
    "C3": (0, 1, 0),      # single-objective: occupancy-oriented
    "C4": (0, 0, 1),      # single-objective: revenue-oriented
    "C5": (1, 1, 1),      # tri-objectives with support-oriented
    "C6": (1, 1, 1),      # tri-objectives with occupancy-oriented
    "C7": (1, 1, 1),      # tri-objectives with revenue-oriented
    "C8": (1, 1, 0),      # bi-objectives: support-oriented and occ
    "C9": (1, 0, 1),      # bi-objectives: support-oriented and rev
    "C10": (0, 1, 1)      # bi-objectives: occupancy-oriented and r
}
current_flags = flags["C"+str(index_case)]

# set of flags for objectives
flag_volume = current_flags[0]
flag_occupancy = current_flags[1]
flag_revenue = current_flags[2]

# Number of top itemsets to consider for TOI mining
k_factor = 5
```

```
The last running = 2026-10-14 18:09:31.135706
Time to load database from .dat: 0.17108702659606934 seconds ---
Load Transaction database successfully
RAM usage after loading DB: 85.20 MB
```

Unique items

Denoted $\mathbf{I}$

$\mathbf{I} \leftarrow scan(\mathbf{D})$

Save $\mathbf{I}$ 'DBname/distinguished_item/distinguished_item_1.dat'

In [2]:
```
# Scan database for distinguished items
import time
import os
#STEP 2: LIST OF DISTINGUISHED ITEMS
import Func_unique
k=1
# Check if the file with distinguished items already exists
file_path = str(dbName)+"/distinguished_item/distinguished_item_"+s
if os.path.exists(file_path):
    # print("File exists")
    with open(file_path, 'r') as file:
        var_unique_items = list(file)
        var_unique_items = [int(line.strip()) for line in var_uniqu
else:
    # print("File does not exist")
    start_time = time.time()
    var_unique_items = Func_unique.Func_unique(DBtrans)
    print("Time to scan and obtain unique items: %s seconds ---" %
    f = open(str(dbName)+"/distinguished_item/distinguished_item_"+
    for item in var_unique_items:
      f.write(str(item) + "\n")
    f.close()
# print('List of distinguished items')
# print(var_unique_items)
print('Number of distinguished items: ', len(var_unique_items))
print(f"RAM usage after obtaining unique items: {memory_usage_mb():

var_unique_1_items = var_unique_items
```

```
Number of distinguished items:  119
RAM usage after obtaining unique items: 85.39 MB
```

Log-normal profit

In [3]:
```
##
import numpy as np
import os
import matplotlib.pyplot as plt

file_path = str(dbName)+"/Profit.dat"
if os.path.exists(file_path):
    print("File exists")
    with open(file_path, 'r') as file:
        var_Profit = list(file)
        var_Profit = [float(line.strip()) for line in var_Profit]
else:
    np.random.seed(42)  # reproducible

    mu = 1.0      # mean of log
    sigma = 0.5   # std of log
```

```
    n = len(var_unique_items)      # number of samples

    Profit = np.random.lognormal(mean=mu, sigma=sigma, size=n)

    var_Profit = (Profit - Profit.min()) / (Profit.max() - Profit.m

    var_Profit = 1 + 9 * var_Profit

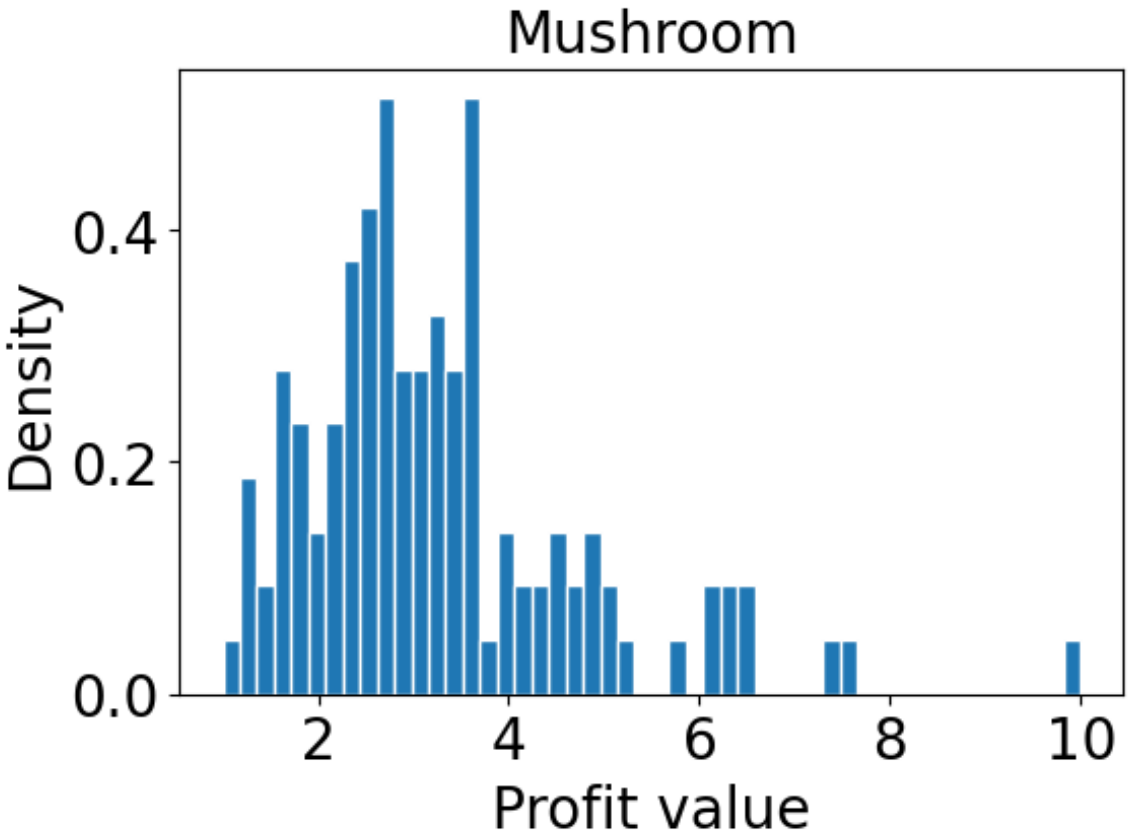
    f = open(file_path, "w")
    for row in var_Profit:
      f.write(str(row)+str("\n"))
    f.close()

print(f"RAM usage after obtaining profit values: {memory_usage_mb()

plt.rcParams.update({
    # "font.family": "serif",
    # "font.serif": ["Times New Roman"],
    "font.size": 20,
    "axes.labelsize": 20,
    "axes.titlesize": 20,
    "xtick.labelsize": 20,
    "ytick.labelsize": 20,
    "legend.fontsize": 20
})

plt.figure(figsize=(6,4))
plt.tight_layout()
plt.hist(var_Profit,
         bins=50,
         density=True,
         edgecolor='white'
         )
plt.xlabel("Profit value")
plt.ylabel("Density")
plt.title(f"{dbName}")
plt.savefig(f"Lognormal_Profit_{dbName}.pdf", dpi=300, bbox_inches=
plt.show()
```

```
File exists
RAM usage after obtaining profit values: 119.89 MB
```



Random volume of item in a transaction

In [4]:
```
import numpy as np
import os

var_Volume = []
file_path = os.path.join(dbName, "Volume.dat")

os.makedirs(dbName, exist_ok=True)

if os.path.exists(file_path):
    print("File exists")

    with open(file_path, 'r', encoding='utf-8') as file:
        var_Volume = [
            list(map(int, line.split()))
            for line in file if line.strip()
        ]

else:
    for transaction in DBtrans:
        n = len(transaction)
        vol = np.random.poisson(lam=3, size=n)
        vol[vol == 0] = 1
        var_Volume.append(
            # np.random.randint(1, 10, size=n).tolist()
            vol.tolist()
        )
```

```
    with open(file_path, "w", encoding='utf-8') as f:
        for row in var_Volume:
            f.write(" ".join(map(str, row)) + "\n")

print(f"RAM usage to random volume values: {memory_usage_mb():.2f} |
```

```
File exists
RAM usage to random volume values: 143.58 MB
```

Statistic length of transactions

$|\mathbf{D}|$

$|\mathbf{T}_i|$

$\min(|\mathbf{T}_i|)$

$\max(|\mathbf{T}_i|)$

$Avg(|\mathbf{T}_i|)$

In [5]:
```
#STEP 3: LENGTH OF TRANSACTION
import statistics

import Func_length_of_transaction
file_path = str(dbName)+"/length_of_transaction_via_volume.dat"
file_path2 = str(dbName)+"/length_of_transaction.dat"
# check if the file with length of transactions already exists
if os.path.exists(file_path) and os.path.exists(file_path2):
    # print("File exists")
    with open(file_path, 'r') as file:
        var_volume_transactions = list(file)
        var_volume_transactions = [int(line.strip()) for line in va
    with open(file_path2, 'r') as file2:
        var_length_transactions = list(file2)
        var_length_transactions = [int(line.strip()) for line in va
else:
    start_time = time.time()
    var_volume_transactions = Func_length_of_transaction.Func_lengt
    var_length_transactions = Func_length_of_transaction.Func_lengt
    print("Time to scan and obtain length of transactions: %s secon
    # save file "length_of_transaction.dat"
    f = open(file_path, "w")
    for length in var_volume_transactions:
      f.write(str(length) + "\n")
    f.close()

    f2 = open(file_path2, "w")
    for length in var_length_transactions:
      f2.write(str(length) + "\n")
    f2.close()
```

```
print('Number of transactions: ', len(var_volume_transactions))
min_transaction_volume = min(var_volume_transactions)
max_transaction_volume = max(var_volume_transactions)
var_mean_transaction_volume = statistics.mean(var_volume_transactio
var_variance_transaction_volume = statistics.variance(var_volume_tr
var_sum_transaction_volume = sum(var_volume_transactions)

print("Max transaction volume: ", max_transaction_volume)
print("Min transaction volume: ", min_transaction_volume)
print("Avg transaction volume: ", var_mean_transaction_volume)
print("Variance transaction volume: ",var_variance_transaction_volu
print("Sum transaction volume: ", var_sum_transaction_volume)
print("Median transaction volume: ", statistics.median(var_volume_t
print("=" * 50)
print("Max transaction length: ", max(var_length_transactions))
print("Min transaction length: ", min(var_length_transactions))
var_mean_length_transaction = statistics.mean(var_length_transactio
print("Avg transaction length: ", var_mean_length_transaction)
print("Variance transaction length: ", statistics.variance(var_leng
print("Sum transaction length: ", sum(var_length_transactions))
median_length_transaction = statistics.median(var_length_transactio
print("Median transaction length: ", median_length_transaction)
max_level = round(np.log2(1+ var_mean_length_transaction * median_l
print("Max level for mining: ", max_level)
rho = (len(var_unique_items) / len(var_length_transactions) * max_l
       + (0 * (len(var_unique_items) / len(var_length_transactions)
)
print("Retention rho =", rho)
print(f"RAM usage after obtaining length of transactions: {memory_u
```

```
Number of transactions:  8124
Max transaction volume:  105
Min transaction volume:  43
Avg transaction volume:  70.17159034958149
Variance transaction volume:  63.610833811730075
Sum transaction volume:  570074
Median transaction volume:  70.0
==================================================
Max transaction length:  23
Min transaction length:  23
Avg transaction length:  23
Variance transaction length:  0
Sum transaction length:  186852
Median transaction length:  23.0
Max level for mining:  9
Retention rho = 0.13183161004431315
RAM usage after obtaining length of transactions: 144.61 MB
```

Dense/ Sparse dataset evaluation

$$\omega = \frac{\sum_{i=1}^{|\mathbf{D}|} |\mathbf{T}_i|}{|\mathbf{D}| \cdot |\mathbf{I}|}$$

In [6]:
```
# Calculate density/sparsity omega
omega = sum(var_length_transactions) / (len(var_length_transactions
print(f"Dense/sparse \u03C9 = {omega}")
```

```
Dense/sparse ω = 0.19327731092436976
```

TDS generation and Robust threshold

$\eta_{\text{sup}}^{(1)}, \mu_{\text{occ}}^{(1)}, \varphi_{\text{pro}}^{(1)}$

In [7]:
```
import gc
import numpy as np
from collections import defaultdict
import json
from pathlib import Path
import time
import matplotlib.pyplot as plt

print(f"RAM usage before generating TDS: {memory_usage_mb():.2f} MB
# ----------------------------
# Step 1: Build TDS
# ----------------------------
profit_dict = {
    item_id: var_Profit[i]
    for i, item_id in enumerate(var_unique_items)
}

start_time = time.time()
TDS = defaultdict(lambda: [set(),[], set(), 0, 0.0, 0.0, 0.0, 0.0,
# [[sorted itemset],[uprofit], {(tid, volume item in tran, len tran

for tid, transaction in enumerate(DBtrans):
    total_volume_of_tran = var_volume_transactions[tid]  # Total vo

    if total_volume_of_tran == 0:
        continue

    for idx, item in enumerate(transaction):
        if item not in profit_dict:
            continue

        key = (item,)
        volume = var_Volume[tid][idx]
        occ = (volume/total_volume_of_tran
               + (
                   (total_volume_of_tran - min_transaction_volume)
                   )
               * (1 - np.exp(-(total_volume_of_tran - volume)/total
        )
        TDS[key][0] = key # sorted itemset
        TDS[key][1] = [profit_dict[item]] # unit profit
        TDS[key][2].add((tid, volume, total_volume_of_tran))
        TDS[key][3] += volume              # volume of item in tran
        TDS[key][4] += occ                # occupancy
```

```
        TDS[key][5] += volume * profit_dict[item] # Revenue of item
# Delete dataset to save memory, free up RAM
DBtrans.clear()
del DBtrans
gc.collect()

volume_vals    = np.array([x[3] for x in TDS.values()])  # volume
occupancy_vals = np.array([x[4] for x in TDS.values()])  # occupanc
revenue_vals   = np.array([x[5] for x in TDS.values()])  # revenue
# print(volume_vals)
# print(occupancy_vals)
# print(revenue_vals)

min_volume = np.min(volume_vals)
print("Min volume   : ", min_volume)
max_volume = np.max(volume_vals)
print("Max volume   : ", max_volume)
min_occupancy = np.min(occupancy_vals)
print("Min occupancy: ", min_occupancy)
max_occupancy = np.max(occupancy_vals)
print("Max occupancy: ", max_occupancy)
min_revenue = np.min(revenue_vals)
print("Min revenue  : ", min_revenue)
max_revenue = np.max(revenue_vals)
print("Max revenue  : ", max_revenue)

normalized_volume = (volume_vals - min_volume) / (max_volume - min_
normalized_occupancy = (occupancy_vals - min_occupancy) / (max_occu
normalized_revenue = (revenue_vals - min_revenue) / (max_revenue -

# items = list(TDS.keys())
# item_to_idx = {item: i for i, item in enumerate(items)}

for i, (item, vals) in enumerate(TDS.items()):
    TDS[item][6] = normalized_volume[i]  # norm volume
    TDS[item][7] = normalized_occupancy[i]
    TDS[item][8] = normalized_revenue[i]
    TDS[item][9] = (
        flag_volume * alpha * normalized_volume[i]
        + flag_occupancy * beta * normalized_occupancy[i]
        + flag_revenue * gamma * normalized_revenue[i]
        )

print("Time to build TDS: %s seconds ---" % (time.time() - start_ti

#################################################
# Drawing sales performance
items_drawing =             [str(item) for item, _ in TDS.items()]
support_vals_drawing =      [vals[3] for _, vals in TDS.items()]
occupancy_vals_drawing =    [vals[4] for _, vals in TDS.items()]
revenue_vals_drawing =      [vals[5] for _, vals in TDS.items()]

if dbName == 'Test':
    x = np.arange(len(items_drawing))
    width = 0.25
```

```
    plt.figure(figsize=(8, 6))

    plt.bar(x - width, support_vals_drawing, width,
            label='Sales-volume performance',
            edgecolor='black',
            color='#1f77b4',
            hatch='///')   # pattern 1

    plt.bar(x, occupancy_vals_drawing, width,
            label='Cross-selling performance',
            edgecolor='black',
            color ='green',
            hatch='xxx')   # pattern 2

    plt.bar(x + width, revenue_vals_drawing, width,
            label='Sales-revenue performance',
            edgecolor='black',
            color='coral',
            hatch='...')   # pattern 3

    # labels = [r'($a$)', r'($b$)', r'($c$)', r'($d$)', r'($e$)', r
    # plt.xticks(x, labels, rotation=0)
    plt.xticks(x, var_unique_items, rotation=0)
    # plt.xticks(x, items_drawing, rotation=45)
    plt.xlabel("Items")
    plt.ylabel("Sales performance")
    plt.title("Volume vs Occupancy vs Revenue")
    plt.legend()
    plt.tight_layout()
    plt.savefig(
        f"Sales_Performance_{dbName}.pdf",
        dpi=300,
        bbox_inches="tight"
    )
    plt.show()

##########################################
## Before sorting by TOI score
# The following code is commented out to save time and memory, sinc
# print("Before sorting by TOI score:")
# for item, vals in TDS.items():  # Print the first 5 items in TDS
#     print(f"""Item: {item}, Unit Profit: {vals[1]}, Support-list:
#           Volume: {vals[3]}, Occupancy: {vals[4]}, Revenue: {vals
#           Norm Volume: {vals[6]}, Norm Occupancy: {vals[7]}, Norm

## After sorting by TOI score
# print("After sorting by TOI score:")
# TDS = dict(sorted(TDS.items(), key=lambda x: x[1][9], reverse=Tru
# for item, vals in TDS.items():
#     print(f"""Item: {item}, Unit Profit: {vals[1]}, Support-list:
#           Support: {vals[3]}, Occupancy: {vals[4]}, Revenue: {val
#           Norm Volume: {vals[6]}, Norm Occupancy: {vals[7]}, Norm

# MAD (Median Absolute Deviation) function
def mad(arr):
    med = np.median(arr)
```

```
    return np.median(np.abs(arr - med))

# Calculate median and MAD for each measure
med_vol, mad_vol = np.median(volume_vals), mad(volume_vals)
med_occ, mad_occ = np.median(occupancy_vals), mad(occupancy_vals)
med_rev, mad_rev = np.median(revenue_vals), mad(revenue_vals)

lambdas = np.arange(-0.5, 1.6, 0.1)

best_lambda_support = None
best_diff = float("inf")

for lam in lambdas:
    theta = med_vol + lam * mad_vol
    ratio = np.sum(volume_vals >= theta) / len(volume_vals)
    diff = abs(ratio - rho)
    if diff < best_diff:
        best_diff = diff
        best_lambda_support = lam
#################
best_lambda_occupancy = None
best_diff = float("inf")

for lam in lambdas:
    theta = med_occ + lam * mad_occ
    ratio = np.sum(occupancy_vals >= theta) / len(occupancy_vals)
    diff = abs(ratio - rho)
    if diff < best_diff:
        best_diff = diff
        best_lambda_occupancy = lam

#################
best_lambda_profit = None
best_diff = float("inf")

for lam in lambdas:
    theta = med_rev + lam * mad_rev
    ratio = np.sum(revenue_vals >= theta) / len(revenue_vals)
    diff = abs(ratio - rho)
    if diff < best_diff:
        best_diff = diff
        best_lambda_profit = lam
# print(best_lambda_occupancy, med_occ + best_lambda_occupancy * ma

# Robust thresholds for support, occupancy, and profit
support_threshold_1   = med_vol + best_lambda_support * mad_vol
occupancy_threshold_1 = med_occ + best_lambda_occupancy * mad_occ
profit_threshold_1    = med_rev + best_lambda_profit * mad_rev

print("Robust thresholds    :")
print(f"Volume threshold    : {support_threshold_1}")
print(f"Occupancy threshold : {occupancy_threshold_1}")
print(f"Revenue threshold   : {profit_threshold_1}")


filtered_TDS = {
```

```
    item: vals
    for item, vals in TDS.items()
    if vals[3] >= flag_volume * support_threshold_1
    and vals[4] >= flag_occupancy * occupancy_threshold_1
    and vals[5] >= flag_revenue * profit_threshold_1
}

# After pre-pruning TDS based on robust thresholds
# print("After pre-pruning TDS based on robust thresholds:")
# for item, vals in filtered_TDS.items():  # Print the first 5 item
#     print(f"""Item: {item}, Unit Profit: {vals[1]}, Support-list:
#           Support: {vals[3]}, Occupancy: {vals[4]}, Revenue: {val
#           Norm Volume: {vals[6]}, Norm Occupancy: {vals[7]}, Norm

top_k_TOI = sorted(filtered_TDS.items(), key=lambda x: x[1][9], rev
print("Top-k TOIs:")
for item, vals in top_k_TOI:
    print(f"""Item: {item}, Unit Profit: {vals[1]}, No. Supports: {
          Volume: {vals[3]}, Occupancy: {vals[4]}, Revenue: {vals[5
          Norm Volume: {vals[6]}, Norm Occupancy: {vals[7]}, Norm R

Total_running_time = time.time() - start_time
print("Time to obtain top-k TOIs: %s seconds ---" % (time.time() -
print(f"RAM after finishing level 1: {memory_usage_mb():.2f} MB")

TOI_nodes = [
    item for item, _ in top_k_TOI
]

set_TOI = set(top_k_TOI[i][0] for i in range(len(top_k_TOI)))
# print(f"Set of top {Top_item_set} items based on TOI mining: {set

avg_bias_support = np.mean([vals[6] for item, vals in top_k_TOI])
avg_bias_occupancy = np.mean([vals[7] for item, vals in top_k_TOI])
avg_bias_revenue = np.mean([vals[8] for item, vals in top_k_TOI])
print(f"Average bias for support among top {k_factor} items    : {a
print(f"Average bias for occupancy among top {k_factor} items  : {a
print(f"Average bias for revenue among top {k_factor} items    : {a
bias_matrix = []
bias_matrix.append([avg_bias_support, avg_bias_occupancy, avg_bias_

# From now on, we can delete the original TDS to save memory,
# since we will work with the filtered TDS for mining TOIs.
# del TDS
# gc.collect()
# print(f"RAM usage after deleting original TDS: {memory_usage_mb()
# Define thresholds for multiple levels
def Sequence_Thresholds(robust_threshold, omega, l_max):
    sequential_thresholds = [robust_threshold]
    factor = 1 / (1 + omega)

    for level in range(1, l_max):
        sequential_thresholds.append((sequential_thresholds[0] * fa

    return sequential_thresholds
```

```
volume_threshold = Sequence_Thresholds(support_threshold_1, omega,
occupancy_threshold = Sequence_Thresholds(occupancy_threshold_1, om
revenue_threshold = Sequence_Thresholds(profit_threshold_1, omega,

print("Sequential thresholds for support    :", [f"{x:.4f}" for x i
print("Sequential thresholds for occupancy  :", [f"{x:.4f}" for x i
print("Sequential thresholds for profit     :", [f"{x:.4f}" for x i
print(f"After pre-pruning, remaining itemsets: {len(filtered_TDS)}"


##################
## Save the filtered TDS to a JSONL file for later use in mining TO
file_path_2 = Path(f"TDS_{dbName}_case_{index_case}.jsonl")

with open(file_path_2, "w", encoding="utf-8") as f:
    for item, vals in filtered_TDS.items():

        record = {
            "item": list(item),
            "sort_itemset": list(item),
            "unitprofit": vals[1],
            "tidset": [
                {
                    "tid": int(tid),
                    "volume": int(vol),
                    "length": int(length)
                }
                for tid, vol, length in vals[2]
            ],

            "volume": int(vals[3]),
            "occupancy": float(vals[4]),
            "revenue": float(vals[5]),
            "norm_volume": float(vals[6]),
            "norm_occupancy": float(vals[7]),
            "norm_revenue": float(vals[8]),
            "combined_score": float(vals[9])
        }

        json.dump(record, f, ensure_ascii=False)
        f.write("\n")

# Draw distributions of support, occupancy, and profit with thresho
plt.rcParams.update({
    "font.size": 18,
    "axes.labelsize": 18,
    "xtick.labelsize": 18,
    "ytick.labelsize": 18
})
# ================================
# Create 1x3 subplots
fig, axes = plt.subplots(1, 3, figsize=(18, 3.5), sharey=True)

# Figure (a) Support
axes[0].hist(np.log10(volume_vals + 1),
             bins=40,
```

```
             density=True,
             edgecolor='white',
            color='#1f77b4',
            hatch='///'
             )
axes[0].axvline(np.log10(support_threshold_1 + 1),
                linestyle='--', linewidth=2, color='red')
# axes[0].axvline(support_threshold_1,
#                 linestyle='--', linewidth=2, color='red')
axes[0].set_xlabel(r"$\log_{10}(Volume(\delta) + 1)$")
axes[0].set_ylabel("Probability Density")

axes[0].text(
    0.5, -0.29, "(a)",
    transform=axes[0].transAxes,
    ha="center", va="top", fontsize=18
)

# Figure (b) Occupancy
axes[1].hist(np.log10(occupancy_vals + 1),
             bins=40,
             density=True,
             edgecolor='white',
             color='green',
             hatch='xxx')
axes[1].axvline(np.log10(occupancy_threshold_1 + 1),
                linestyle='--', linewidth=2, color='red')
# axes[1].axvline(occupancy_threshold_1,
#                 linestyle='--', linewidth=2, color='red')
axes[1].set_xlabel(r"$\log_{10}(Occupancy(\delta) + 1)$")

axes[1].text(
    0.5, -0.29, "(b)",
    transform=axes[1].transAxes,
    ha="center", va="top", fontsize=18
)

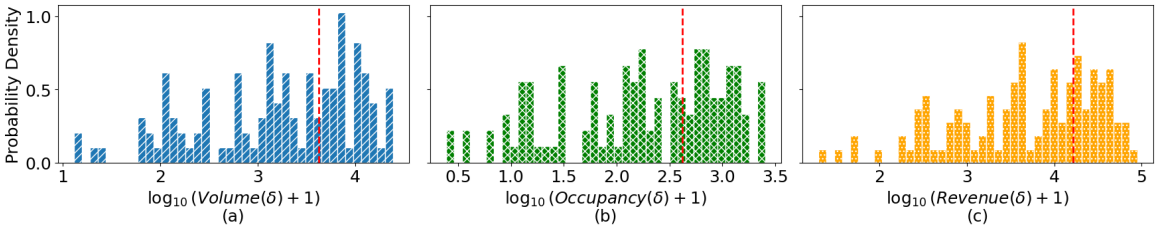
# Figure (c) Revenue
axes[2].hist(np.log10(revenue_vals + 1),
             bins=40,
             density=True,
             edgecolor='white',
             color='orange',
             hatch='...')
axes[2].axvline(np.log10(profit_threshold_1 + 1),
                linestyle='--', linewidth=2, color='red')
# axes[2].axvline(profit_threshold_1,
#                 linestyle='--', linewidth=2, color='red')
axes[2].set_xlabel(r"$\log_{10}(Revenue(\delta) + 1)$")

axes[2].text(
    0.5, -0.29, "(c)",
    transform=axes[2].transAxes,
    ha="center", va="top", fontsize=18
)
```

```
plt.tight_layout()
plt.subplots_adjust(bottom=0.25)  # Adjust bottom to fit x-labels
plt.savefig(
    f"Robust_Threshold_Distributions_{dbName}.pdf",
    dpi=300,
    bbox_inches="tight"
)
plt.show()
```

```
RAM usage before generating TDS: 146.34 MB
Min volume   :  12
Max volume   :  24541
Min occupancy:  1.4466344401718074
Max occupancy:  2542.014063553579
Min revenue  :  19.0
Max revenue  :  89636.80026070864
Time to build TDS: 0.1830441951751709 seconds ---
Robust thresholds    :
Volume threshold    : 4218.999999999999
Occupancy threshold : 418.56599704873645
Revenue threshold   : 16412.909620371152
Top-k TOIs:
Item: (90,), Unit Profit: [3.915638662445506], No. Supports: 7488,
          Volume: 22892, Occupancy: 2345.5081124251997, Revenue: 896
36.80026070864,
          Norm Volume: 0.9327734518325248, Norm Occupancy: 0.9226527
314817403, Norm Revenue: 1.0, TOI Score: 0.9663867259162624
Item: (86,), Unit Profit: [2.4489926533766697], No. Supports: 7924,
          Volume: 24378, Occupancy: 2483.8704688052085, Revenue: 597
01.542904016176,
          Norm Volume: 0.9933548045171022, Norm Occupancy: 0.9771139
336503809, Norm Revenue: 0.6659675056784778, TOI Score: 0.8296611550
9779
Item: (85,), Unit Profit: [2.133578910892262], No. Supports: 8124,
          Volume: 24541, Occupancy: 2542.014063553579, Revenue: 5236
0.16005221159,
          Norm Volume: 1.0, Norm Occupancy: 1.0, Norm Revenue: 0.584
0487034935588, TOI Score: 0.7920243517467793
Item: (34,), Unit Profit: [1.9106395472365856], No. Supports: 7914,
          Volume: 24182, Occupancy: 2479.2415104320667, Revenue: 462
03.085531273726,
          Norm Volume: 0.985364262709446, Norm Occupancy: 0.97529191
61278005, Norm Revenue: 0.5153450028556696, TOI Score: 0.75035463278
25579
Item: (59,), Unit Profit: [3.5951056801831824], No. Supports: 5176,
          Volume: 15840, Occupancy: 1628.4204238342215, Revenue: 569
46.473974100234,
          Norm Volume: 0.645277019038689, Norm Occupancy: 0.64039779
88341847, Norm Revenue: 0.6352250759167439, TOI Score: 0.64025104747
77164
Time to obtain top-k TOIs: 0.18397307395935059 seconds ---
RAM after finishing level 1: 168.19 MB
Average bias for support among top 5 items    : 0.9114
Average bias for occupancy among top 5 items  : 0.9031
Average bias for revenue among top 5 items    : 0.6801
Sequential thresholds for support    : ['4219.0000', '1767.8204', '1
```

```
178.5469', '883.9102', '707.1282', '589.2735', '505.0915', '441.955
1', '392.8490']
Sequential thresholds for occupancy  : ['418.5660', '175.3850', '11
6.9234', '87.6925', '70.1540', '58.4617', '50.1100', '43.8463', '38.
9745']
Sequential thresholds for profit     : ['16412.9096', '6877.2403', '
4584.8269', '3438.6201', '2750.8961', '2292.4134', '1964.9258', '171
9.3101', '1528.2756']
After pre-pruning, remaining itemsets: 38
```



Baseline support calculation of single itemsets

$$\mathcal{S}up(\mathbf{S}) \leftarrow \mathcal{S}up(\mathbf{S}) \cup Tid, s.t. \quad \mathbf{T}_i \supseteq \mathbf{S}, i = 1, \ldots, |\mathbf{D}|.$$

and

$$|\mathcal{S}up\,(\mathbf{S})| = \sum_{i=1}^{|\mathbf{D}|} I, s.t. \quad \begin{bmatrix} I=1 | \mathbf{S} \subseteq \mathbf{T}_i \\ I=0 | \mathbf{S} \nsubseteq \mathbf{T}_i \end{bmatrix}.$$

Baseline Occupancy

$$Occupancy\,(\mathbf{S}) = \sum_{\mathbf{S} \in \mathbf{T}_i} \frac{|\mathbf{S}|}{|\mathbf{T}_i|}$$

Length-rewarded Occupancy

$$\mathcal{O}cc\,(\mathbf{S}) = \sum\nolimits_{\forall \mathbf{T}_i \in \mathcal{S}up(\mathbf{S})} \frac{|\mathbf{S}|}{|\mathbf{T}_i|} + \left(1 - \exp\left(-\frac{|\mathbf{T}_i| - |\mathbf{S}|}{|\mathbf{T}_i|}\right)\right)$$

profit calculation of a single item (level-wise $\ell = 1$)

$$n\mathcal{P}ro = |\mathcal{S}up(\mathbf{S})| \cdot u\mathcal{P}ro(\mathbf{S})$$

Level-wise TOI mining

$$TOI(\mathbf{S}) = \alpha \cdot \mathcal{S}up(\mathbf{S}) + \beta \cdot \mathcal{O}cc(\mathbf{S}) + \gamma \cdot n\mathcal{P}ro(\mathbf{S})$$

where:

1. $\mathcal{S}up(\mathbf{S}) \geq \eta_{\text{sup}}$.
2. $\mathcal{O}cc(\mathbf{S}) \geq \mu_{\text{occ}}$.
3. $n\mathcal{P}ro(\mathbf{S}) \geq \varphi_{\text{pro}}$

In [8]:
```
import numpy as np
import time
from collections import defaultdict
import heapq
import json
import os

# ==================== Encode TIDs ====================
all_tids = set()
for vals in TDS.values(): # filtered_TDS
    for tid, _, _ in vals[2]:
        all_tids.add(tid)

tid_to_idx = {tid: i for i, tid in enumerate(sorted(all_tids))}
idx_to_tid = {i: tid for tid, i in tid_to_idx.items()}
N = len(tid_to_idx)

# ==================== STEP 2: Build bitset + data ===============
bitset_dict = {}
data_dict = {}

for item, vals in TDS.items(): # filtered_TDS
    bitset = np.zeros(N, dtype=np.bool_)
    data = {}

    for tid, vol, l in vals[2]:
        idx = tid_to_idx[tid]
        bitset[idx] = True
        data[idx] = (vol if isinstance(vol, list) else [vol], l)

    bitset_dict[item] = bitset
    data_dict[item] = data

# ==================== INIT ====================
current_level = 2
next_top_k_TOI = dict(top_k_TOI)

bitset_cache = {}
data_cache = {}

# init cache từ level 1
for item, vals in next_top_k_TOI.items():
```

```
    base_item = item if isinstance(item, tuple) else (item,)
    b = np.zeros(N, dtype=np.bool_)
    d = {}

    for tid, vol, l in vals[2]:
        idx = tid_to_idx[tid]
        b[idx] = True
        d[idx] = (vol if isinstance(vol, list) else [vol], l)

    bitset_cache[base_item] = b
    data_cache[base_item] = d

# ==================== MAIN LOOP ====================
while next_top_k_TOI and current_level <= max_level:
    start_time = time.time()
    seen_itemsets = set()

    current_top_k_TOI = defaultdict(
        lambda: [set(), [], None, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0
    )

    for itemset, values in next_top_k_TOI.items():
        base_items = itemset if isinstance(itemset, tuple) else (it

        bitA = bitset_cache[base_items]
        dataA = data_cache[base_items]

        for item, value in TDS.items(): # filtered_TDS
            if item[0] in base_items:
                continue

            new_items = base_items + item
            key = frozenset(new_items)

            if key in seen_itemsets:
                continue
            seen_itemsets.add(key)

            bitB = bitset_dict[item]
            dataB = data_dict[item]

            # BITSET INTERSECTION
            common_bits = bitA & bitB

            if not common_bits.any():
                continue

            common_idx = np.nonzero(common_bits)[0]

            list_of_supports = []
            append = list_of_supports.append

            for idx in common_idx:
                volA, lenTid = dataA[idx]
                volB, _ = dataB[idx]
```

```
                new_vol = volB + volA
                append((idx, new_vol, lenTid))

            if not list_of_supports:
                continue

            profit_itemset = values[1] + value[1]

            # ===================== METRICS =====================
            volume = 0.0
            occupancy = 0.0
            revenue = 0.0

            for _, vol_itemset, lenTid in list_of_supports:
                s = sum(vol_itemset)
                volume += s
                occupancy += (s / lenTid
                              + (
                                  (lenTid - min_transaction_volume)
                                )
                               * (1 - np.exp(-(lenTid - s) / lenTid
                )

                for j in range(len(vol_itemset)):
                    revenue += profit_itemset[j] * vol_itemset[j]

            if (
                volume < flag_volume * volume_threshold[current_lev
                or occupancy < flag_occupancy * occupancy_threshold
                or revenue < flag_revenue * revenue_threshold[curre
            ):
                continue

            # CACHE BITSET + DATA
            new_data = {
                idx: (vol, lenTid)
                for idx, vol, lenTid in list_of_supports
            }

            bitset_cache[new_items] = common_bits
            data_cache[new_items] = new_data

            current_top_k_TOI[new_items][0] = new_items
            current_top_k_TOI[new_items][1] = profit_itemset
            current_top_k_TOI[new_items][2] = list_of_supports
            current_top_k_TOI[new_items][3] = volume
            current_top_k_TOI[new_items][4] = occupancy
            current_top_k_TOI[new_items][5] = revenue

    if not current_top_k_TOI:
        break

    # ===================== NORMALIZE =====================
    volumes = [vals[3] for vals in current_top_k_TOI.values()]
    occupancies = [vals[4] for vals in current_top_k_TOI.values()]
    revenues = [vals[5] for vals in current_top_k_TOI.values()]
```

```
    min_volume, max_volume = min(volumes), max(volumes)
    min_occupancy, max_occupancy = min(occupancies), max(occupancie
    min_revenue, max_revenue = min(revenues), max(revenues)

    for item, vals in current_top_k_TOI.items():
        vals[6] = (vals[3] - min_volume) / (max_volume - min_volume
        vals[7] = (vals[4] - min_occupancy) / (max_occupancy - min_
        vals[8] = (vals[5] - min_revenue) / (max_revenue - min_reve
        vals[9] = (flag_volume * alpha * vals[6]
                   + flag_occupancy * beta * vals[7]
                   + flag_revenue * gamma * vals[8]
        )


    # TOP-K
    current_top_k_TOI = dict(heapq.nlargest(
        k_factor,
        current_top_k_TOI.items(),
        key=lambda x: x[1][9]
    ))

    # ==================== OUTPUT ====================
    print("-" * 50)
    for item, vals in current_top_k_TOI.items():
        print(f"""Item: {item}, Unit Profit: {vals[1]}, No. Support
Volume: {vals[3]}, Occupancy: {vals[4]}, Revenue: {vals[5]},
Norm Volume: {vals[6]}, Norm Occupancy: {vals[7]}, Norm Revenue: {v

    TOI_nodes.extend([tuple(item) for item in current_top_k_TOI.key

    avg_bias_support = np.mean([vals[6] for vals in current_top_k_T
    avg_bias_occupancy = np.mean([vals[7] for vals in current_top_k
    avg_bias_revenue = np.mean([vals[8] for vals in current_top_k_T

    print(f"Average bias support   : {avg_bias_support:.4f}")
    print(f"Average bias occupancy : {avg_bias_occupancy:.4f}")
    print(f"Average bias revenue   : {avg_bias_revenue:.4f}")

    bias_matrix.append([avg_bias_support, avg_bias_occupancy, avg_b

    Total_running_time += time.time() - start_time
    print("Time:", time.time() - start_time)
    print(f"RAM level {current_level}: {memory_usage_mb():.2f} MB")

    next_top_k_TOI = current_top_k_TOI

    for r in next_top_k_TOI:
        set_TOI.add(r)

    current_level += 1

# ==================== FINAL ====================
print("=" * 50)
avg_bias_weights = np.mean(bias_matrix, axis=0)
print(f"Support  : {avg_bias_weights[0]:.4f}")
```

```
print(f"Occupancy: {avg_bias_weights[1]:.4f}")
print(f"Revenue  : {avg_bias_weights[2]:.4f}")
print("=" * 50)
print(f"Total running time: {Total_running_time:.4f} seconds")
print(f"Final TOI: {set_TOI}")
print("=" * 50)
# ====================== SAVE ======================
file_path = f"Average_Bias_Weights_{dbName}.json"
key = "C" + str(index_case)
value = avg_bias_weights.tolist()

if os.path.exists(file_path):
    with open(file_path, "r") as f:
        data = json.load(f)
else:
    data = {}

data[key] = value

with open(file_path, "w") as f:
    json.dump(data, f, indent=4)
```

```
--------------------------------------------------
Item: (90, 86), Unit Profit: [3.915638662445506, 2.448992653376669
7], No. Supports: 7288,
Volume: 44646.0, Occupancy: 2545.594346212228, Revenue: 142156.86374
759898,
Norm Volume: 0.9179762771378409, Norm Occupancy: 0.9173882114623129,
Norm Revenue: 1.0, TOI Score: 0.9589881385689205
Item: (90, 85), Unit Profit: [3.915638662445506, 2.133578910892262],
No. Supports: 7488,
Volume: 45457.0, Occupancy: 2611.205133307328, Revenue: 137198.27484
623223,
Norm Volume: 0.9353402132488331, Norm Occupancy: 0.9420171536333567,
Norm Revenue: 0.9633384053146676, TOI Score: 0.9493393092817504
Item: (90, 34), Unit Profit: [3.915638662445506, 1.910639547236585
6], No. Supports: 7296,
Volume: 44596.0, Occupancy: 2547.7279659233195, Revenue: 129904.3265
2392489,
Norm Volume: 0.9169057508671263, Norm Occupancy: 0.9181891284884004,
Norm Revenue: 0.9094102046980596, TOI Score: 0.9131579777825929
Item: (86, 85), Unit Profit: [2.4489926533766697, 2.13357891089226
2], No. Supports: 7924,
Volume: 48309.0, Occupancy: 2765.669753091219, Revenue: 110619.22987
767706,
Norm Volume: 0.9964030317303987, Norm Occupancy: 1.0, Norm Revenue:
0.7668248017183281, TOI Score: 0.8816139167243634
Item: (86, 34), Unit Profit: [2.4489926533766697, 1.910639547236585
6], No. Supports: 7906,
Volume: 48477.0, Occupancy: 2762.7847802701453, Revenue: 105627.0693
1640758,
Norm Volume: 1.0, Norm Occupancy: 0.9989170404453118, Norm Revenue:
0.7299149931534527, TOI Score: 0.8649574965767264
Average bias support   : 0.9533
Average bias occupancy : 0.9553
Average bias revenue   : 0.8739
```

```
Time: 1.045928955078125
RAM level 2: 416.55 MB
--------------------------------------------------
Item: (90, 86, 85), Unit Profit: [3.915638662445506, 2.4489926533766
697, 2.133578910892262], No. Supports: 7288,
Volume: 66601.0, Occupancy: 2801.8342831457176, Revenue: 188281.6242
052642,
Norm Volume: 0.9185679285097222, Norm Occupancy: 0.919299670780757,
Norm Revenue: 1.0, TOI Score: 0.9592839642548612
Item: (90, 86, 34), Unit Profit: [3.915638662445506, 2.4489926533766
697, 1.9106395472365856], No. Supports: 7288,
Volume: 66914.0, Occupancy: 2805.739763278243, Revenue: 184542.58241
737218,
Norm Volume: 0.9230006656186713, Norm Occupancy: 0.9206156057966952,
Norm Revenue: 0.9796382319374898, TOI Score: 0.9513194487780805
Item: (90, 85, 34), Unit Profit: [3.915638662445506, 2.1335789108922
62, 1.9106395472365856], No. Supports: 7296,
Volume: 66572.0, Occupancy: 2804.225141574477, Revenue: 176791.85666
96859,
Norm Volume: 0.9181572276274235, Norm Occupancy: 0.9201052604183417,
Norm Revenue: 0.9374299633640893, TOI Score: 0.9277935954957564
Item: (86, 85, 34), Unit Profit: [2.4489926533766697, 2.133578910892
262, 1.9106395472365856], No. Supports: 7906,
Volume: 72351.0, Occupancy: 3041.3396756132383, Revenue: 156564.1322
3503952,
Norm Volume: 1.0, Norm Occupancy: 1.0, Norm Revenue: 0.8272754754671
134, TOI Score: 0.9136377377335567
Item: (90, 85, 36), Unit Profit: [3.915638662445506, 2.1335789108922
62, 1.779050812031834], No. Supports: 6464,
Volume: 59349.0, Occupancy: 2491.8104305279, Revenue: 155268.4472569
318,
Norm Volume: 0.815864383736245, Norm Occupancy: 0.8148384446082867,
Norm Revenue: 0.820219540149209, TOI Score: 0.818041961942727
Average bias support   : 0.9151
Average bias occupancy : 0.9150
Average bias revenue   : 0.9129
Time: 0.9724080562591553
RAM level 3: 594.42 MB
--------------------------------------------------
Item: (90, 86, 85, 34), Unit Profit: [3.915638662445506, 2.448992653
3766697, 2.133578910892262, 1.9106395472365856], No. Supports: 7288,
Volume: 88869.0, Occupancy: 3059.49605073448, Revenue: 231252.202806
67586,
Norm Volume: 1.0, Norm Occupancy: 1.0, Norm Revenue: 1.0, TOI Score:
1.0
Item: (90, 86, 85, 36), Unit Profit: [3.915638662445506, 2.448992653
3766697, 2.133578910892262, 1.779050812031834], No. Supports: 6272,
Volume: 76750.0, Occupancy: 2638.510890679697, Revenue: 197463.03106
024003,
Norm Volume: 0.8617657123303296, Norm Occupancy: 0.8605279500146744,
Norm Revenue: 0.8514666300667536, TOI Score: 0.8566161711985416
Item: (90, 86, 34, 36), Unit Profit: [3.915638662445506, 2.448992653
3766697, 1.9106395472365856, 1.779050812031834], No. Supports: 6272,
Volume: 77078.0, Occupancy: 2642.493656429942, Revenue: 193987.20250
455037,
Norm Volume: 0.8655070149423977, Norm Occupancy: 0.8618474371522393,
```

```
Norm Revenue: 0.8361872816245636, TOI Score: 0.8508471482834806
Item: (90, 85, 34, 36), Unit Profit: [3.915638662445506, 2.133578910
892262, 1.9106395472365856, 1.779050812031834], No. Supports: 6272,
Volume: 76765.0, Occupancy: 2639.07840831215, Revenue: 187330.389746
87233,
Norm Volume: 0.8619368084863693, Norm Occupancy: 0.8607159681566695,
Norm Revenue: 0.8069246871066367, TOI Score: 0.834430747796503
Item: (86, 85, 34, 36), Unit Profit: [2.4489926533766697, 2.13357891
0892262, 1.9106395472365856, 1.779050812031834], No. Supports: 6602,
Volume: 80779.0, Occupancy: 2777.419933238415, Revenue: 167175.80210
468368,
Norm Volume: 0.9077221398425915, Norm Occupancy: 0.9065484057586302,
Norm Revenue: 0.7183274037942684, TOI Score: 0.81302477181843
Average bias support   : 0.8994
Average bias occupancy : 0.8979
Average bias revenue   : 0.8426
Time: 1.2093658447265625
RAM level 4: 746.55 MB
------------------------------------------------
Item: (90, 86, 85, 34, 36), Unit Profit: [3.915638662445506, 2.44899
26533766697, 2.133578910892262, 1.9106395472365856, 1.77905081203183
4], No. Supports: 6272,
Volume: 95959.0, Occupancy: 2858.147701229821, Revenue: 234271.30592
109953,
Norm Volume: 1.0, Norm Occupancy: 1.0, Norm Revenue: 1.0, TOI Score:
1.0
Item: (90, 86, 85, 34, 59), Unit Profit: [3.915638662445506, 2.44899
26533766697, 2.133578910892262, 1.9106395472365856, 3.59510568018318
24], No. Supports: 4544,
Volume: 69557.0, Occupancy: 2073.500690927486, Revenue: 195001.10674
656022,
Norm Volume: 0.7206048869276274, Norm Occupancy: 0.7212528207768071,
Norm Revenue: 0.8300567378618869, TOI Score: 0.7753308123947571
Item: (90, 86, 85, 34, 39), Unit Profit: [3.915638662445506, 2.44899
26533766697, 2.133578910892262, 1.9106395472365856, 1.69813857538314
08], No. Supports: 4784,
Volume: 72819.0, Occupancy: 2172.9920100960376, Revenue: 176010.3413
3375582,
Norm Volume: 0.7551245013069198, Norm Occupancy: 0.7565972804017055,
Norm Revenue: 0.7478734871376564, TOI Score: 0.751498994222288
Item: (90, 86, 85, 34, 53), Unit Profit: [3.915638662445506, 2.44899
26533766697, 2.133578910892262, 1.9106395472365856, 2.26297218235440
53], No. Supports: 4608,
Volume: 70042.0, Occupancy: 2093.6470730892283, Revenue: 177498.0381
314304,
Norm Volume: 0.7257373249944443, Norm Occupancy: 0.7284098571580128,
Norm Revenue: 0.7543115509158804, TOI Score: 0.7400244379551624
Item: (90, 86, 85, 34, 63), Unit Profit: [3.915638662445506, 2.44899
26533766697, 2.133578910892262, 1.9106395472365856, 1.87028879362762
87], No. Supports: 4304,
Volume: 65508.0, Occupancy: 1956.4093028744521, Revenue: 160941.3088
226229,
Norm Volume: 0.6777569658296031, Norm Occupancy: 0.6796559068746751,
Norm Revenue: 0.6826616828678287, TOI Score: 0.680209324348716
Average bias support   : 0.7758
Average bias occupancy : 0.7772
```

```
Average bias revenue   : 0.8030
Time: 0.9009678363800049
RAM level 5: 889.69 MB
--------------------------------------------------
Item: (90, 86, 85, 34, 36, 59), Unit Profit: [3.915638662445506, 2.4
489926533766697, 2.133578910892262, 1.9106395472365856, 1.7790508120
31834, 3.5951056801831824], No. Supports: 3936,
Volume: 72449.0, Occupancy: 1934.0945751092934, Revenue: 190809.7023
1464086,
Norm Volume: 0.983950821024837, Norm Occupancy: 0.9830923316234054,
Norm Revenue: 1.0, TOI Score: 0.9919754105124186
Item: (90, 86, 85, 34, 36, 39), Unit Profit: [3.915638662445506, 2.4
489926533766697, 2.133578910892262, 1.9106395472365856, 1.7790508120
31834, 1.6981385753831408], No. Supports: 4016,
Volume: 73616.0, Occupancy: 1966.9532206369856, Revenue: 170091.0890
7423567,
Norm Volume: 1.0, Norm Occupancy: 1.0, Norm Revenue: 0.8900721390188
067, TOI Score: 0.9450360695094033
Item: (90, 86, 85, 34, 36, 1), Unit Profit: [3.915638662445506, 2.44
89926533766697, 2.133578910892262, 1.9106395472365856, 1.77905081203
1834, 3.8852654429379183], No. Supports: 3696,
Volume: 67399.0, Occupancy: 1806.3422233804997, Revenue: 180379.5343
6876278,
Norm Volume: 0.9145006463679621, Norm Occupancy: 0.9173563780357827,
Norm Revenue: 0.944660096761256, TOI Score: 0.929580371564609
Item: (90, 86, 85, 34, 36, 110), Unit Profit: [3.915638662445506, 2.
4489926533766697, 2.133578910892262, 1.9106395472365856, 1.779050812
031834, 2.977386383720558], No. Supports: 3664,
Volume: 67470.0, Occupancy: 1798.9618394242527, Revenue: 170910.4899
2369487,
Norm Volume: 0.9154770745661083, Norm Occupancy: 0.9135587447974012,
Norm Revenue: 0.8944196781915327, TOI Score: 0.9049483763788204
Item: (90, 86, 85, 34, 59, 63), Unit Profit: [3.915638662445506, 2.4
489926533766697, 2.133578910892262, 1.9106395472365856, 3.5951056801
831824, 1.8702887936276287], No. Supports: 3596,
Volume: 65990.0, Occupancy: 1764.130817787182, Revenue: 174544.45563
75337,
Norm Volume: 0.8951233600132024, Norm Occupancy: 0.8956361751479228,
Norm Revenue: 0.9137006055956047, TOI Score: 0.9044119828044035
Average bias support   : 0.9418
Average bias occupancy : 0.9419
Average bias revenue   : 0.9286
Time: 0.49053382873535156
RAM level 6: 978.39 MB
--------------------------------------------------
Item: (90, 86, 85, 34, 36, 39, 56), Unit Profit: [3.915638662445506,
2.4489926533766697, 2.133578910892262, 1.9106395472365856, 1.7790508
12031834, 1.6981385753831408, 4.772217976379022], No. Supports: 331
2,
Volume: 70658.0, Occupancy: 1733.4795536061679, Revenue: 188501.1154
1616468,
Norm Volume: 1.0, Norm Occupancy: 1.0, Norm Revenue: 1.0, TOI Score:
1.0
Item: (90, 86, 85, 34, 36, 59, 63), Unit Profit: [3.915638662445506,
2.4489926533766697, 2.133578910892262, 1.9106395472365856, 1.7790508
12031834, 3.5951056801831824, 1.8702887936276287], No. Supports: 319
```

```
2,
Volume: 68468.0, Occupancy: 1676.7452704485502, Revenue: 172615.5476
8823646,
Norm Volume: 0.9687048971834408, Norm Occupancy: 0.9669536684225595,
Norm Revenue: 0.9148348975777418, TOI Score: 0.9417698973805912
Item: (90, 86, 85, 34, 36, 1, 24), Unit Profit: [3.915638662445506,
2.4489926533766697, 2.133578910892262, 1.9106395472365856, 1.7790508
12031834, 3.8852654429379183, 1.628971218190871], No. Supports: 315
2,
Volume: 66947.0, Occupancy: 1644.693063509984, Revenue: 169293.17779
885203,
Norm Volume: 0.9469698052272825, Norm Occupancy: 0.9482840422401341,
Norm Revenue: 0.8970231342992661, TOI Score: 0.9219964697632743
Item: (90, 86, 85, 34, 36, 59, 93), Unit Profit: [3.915638662445506,
2.4489926533766697, 2.133578910892262, 1.9106395472365856, 1.7790508
12031834, 3.5951056801831824, 2.2375950763560497], No. Supports: 302
4,
Volume: 65021.0, Occupancy: 1591.2372615698794, Revenue: 167335.2486
3532785,
Norm Volume: 0.9194472627502537, Norm Occupancy: 0.9171473457431853,
Norm Revenue: 0.8865263588233788, TOI Score: 0.9029868107868162
Item: (90, 86, 85, 34, 59, 63, 93), Unit Profit: [3.915638662445506,
2.4489926533766697, 2.133578910892262, 1.9106395472365856, 3.5951056
801831824, 1.8702887936276287, 2.2375950763560497], No. Supports: 29
60,
Volume: 63595.0, Occupancy: 1556.2194956127435, Revenue: 164570.9498
914917,
Norm Volume: 0.8990697209162749, Norm Occupancy: 0.8967503540035383,
Norm Revenue: 0.8717065053767417, TOI Score: 0.8853881131465082
Average bias support   : 0.9468
Average bias occupancy : 0.9458
Average bias revenue   : 0.9140
Time: 0.6507940292358398
RAM level 7: 1052.06 MB
--------------------------------------------------
Item: (90, 86, 85, 34, 36, 59, 93, 23), Unit Profit: [3.915638662445
506, 2.4489926533766697, 2.133578910892262, 1.9106395472365856, 1.77
9050812031834, 3.5951056801831824, 2.2375950763560497, 3.17945434355
13303], No. Supports: 2832,
Volume: 69560.0, Occupancy: 1585.4026361142073, Revenue: 184349.5085
500069,
Norm Volume: 1.0, Norm Occupancy: 1.0, Norm Revenue: 1.0, TOI Score:
1.0
Item: (90, 86, 85, 34, 59, 63, 93, 23), Unit Profit: [3.915638662445
506, 2.4489926533766697, 2.133578910892262, 1.9106395472365856, 3.59
51056801831824, 1.8702887936276287, 2.2375950763560497, 3.1794543435
513303], No. Supports: 2672,
Volume: 65686.0, Occupancy: 1496.0375353710542, Revenue: 174789.0457
9502268,
Norm Volume: 0.9437024980744917, Norm Occupancy: 0.9430006553506955,
Norm Revenue: 0.9476155594665369, TOI Score: 0.9456590287705142
Item: (90, 86, 85, 34, 36, 59, 63, 93), Unit Profit: [3.915638662445
506, 2.4489926533766697, 2.133578910892262, 1.9106395472365856, 1.77
9050812031834, 3.5951056801831824, 1.8702887936276287, 2.23759507635
60497], No. Supports: 2760,
Volume: 67797.0, Occupancy: 1545.2439415615606, Revenue: 168548.1652
```

0358392,
Norm Volume: 0.9743798410184122, Norm Occupancy: 0.9743857585071001,
Norm Revenue: 0.9134200349823031, TOI Score: 0.9438999380003577
Item: (90, 86, 85, 34, 36, 59, 63, 23), Unit Profit: [3.915638662445
506, 2.4489926533766697, 2.133578910892262, 1.9106395472365856, 1.77
9050812031834, 3.5951056801831824, 1.8702887936276287, 3.17945434355
13303], No. Supports: 2568,
Volume: 63116.0, Occupancy: 1437.7805701692696, Revenue: 164429.5674
0958124,
Norm Volume: 0.9063549038699084, Norm Occupancy: 0.9058428744032659,
Norm Revenue: 0.89085308955817, TOI Score: 0.8986039967140391
Item: (90, 86, 85, 34, 36, 59, 93, 2), Unit Profit: [3.9156386624455
06, 2.4489926533766697, 2.133578910892262, 1.9106395472365856, 1.779
050812031834, 3.5951056801831824, 2.2375950763560497, 2.891124748491
2592], No. Supports: 2528,
Volume: 61954.0, Occupancy: 1412.6841700946602, Revenue: 161799.9206
8460185,
Norm Volume: 0.8894685597198204, Norm Occupancy: 0.8898357491898018,
Norm Revenue: 0.8764445221050011, TOI Score: 0.8829565409124107
Average bias support   : 0.9428
Average bias occupancy : 0.9426
Average bias revenue   : 0.9257
Time: 0.3311429023742676
RAM level 8: 1113.06 MB

Item: (90, 86, 85, 34, 36, 59, 93, 23, 63), Unit Profit: [3.91563866
2445506, 2.4489926533766697, 2.133578910892262, 1.9106395472365856,
1.779050812031834, 3.5951056801831824, 2.2375950763560497, 3.1794543
435513303, 1.8702887936276287], No. Supports: 2568,
Volume: 70996.0, Occupancy: 1523.516117376579, Revenue: 181863.20656
47784,
Norm Volume: 0.9995453705940017, Norm Occupancy: 0.9983900881652892,
Norm Revenue: 1.0, TOI Score: 0.9997726852970008
Item: (90, 86, 85, 34, 36, 59, 93, 23, 39), Unit Profit: [3.91563866
2445506, 2.4489926533766697, 2.133578910892262, 1.9106395472365856,
1.779050812031834, 3.5951056801831824, 2.2375950763560497, 3.1794543
435513303, 1.6981385753831408], No. Supports: 2576,
Volume: 71028.0, Occupancy: 1525.950196134113, Revenue: 180464.22073
755378,
Norm Volume: 1.0, Norm Occupancy: 1.0, Norm Revenue: 0.9922417138109
665, TOI Score: 0.9961208569054832
Item: (90, 86, 85, 34, 36, 59, 93, 23, 2), Unit Profit: [3.915638662
445506, 2.4489926533766697, 2.133578910892262, 1.9106395472365856,
1.779050812031834, 3.5951056801831824, 2.2375950763560497, 3.1794543
435513303, 2.8911247484912592], No. Supports: 2432,
Volume: 66950.0, Occupancy: 1439.1535289232238, Revenue: 179009.1993
7213592,
Norm Volume: 0.9420631650730958, Norm Occupancy: 0.9425922512475189,
Norm Revenue: 0.9841726741220763, TOI Score: 0.9631179195975861
Item: (90, 86, 85, 34, 36, 59, 93, 2, 39), Unit Profit: [3.915638662
445506, 2.4489926533766697, 2.133578910892262, 1.9106395472365856,
1.779050812031834, 3.5951056801831824, 2.2375950763560497, 2.8911247
484912592, 1.6981385753831408], No. Supports: 2432,
Volume: 66958.0, Occupancy: 1438.9754110697083, Revenue: 168017.9762
8675797,
Norm Volume: 0.9421768224245954, Norm Occupancy: 0.942474443210511,

```
Norm Revenue: 0.9232191943909459, TOI Score: 0.9326980084077707
Item: (90, 86, 85, 34, 59, 63, 93, 23, 2), Unit Profit: [3.915638662
445506, 2.4489926533766697, 2.133578910892262, 1.9106395472365856,
3.5951056801831824, 1.8702887936276287, 2.2375950763560497, 3.179454
3435513303, 2.8911247484912592], No. Supports: 2336,
Volume: 64456.0, Occupancy: 1383.9123556957036, Revenue: 173133.2896
9621786,
Norm Volume: 0.906630485743106, Norm Occupancy: 0.9060554636571319,
Norm Revenue: 0.9515869337953702, TOI Score: 0.9291087097692381
Average bias support   : 0.9581
Average bias occupancy : 0.9579
Average bias revenue   : 0.9702
Time: 0.6101040840148926
RAM level 9: 1162.44 MB
==================================================
Support  : 0.9161
Occupancy: 0.9152
Revenue  : 0.8723
==================================================
Total running time: 6.3952 seconds
Final TOI: {(90, 85), (86, 85, 34, 36), (90, 86, 85, 34, 36, 59, 93,
2, 39), (90, 85, 36), (90, 86, 34, 36), (90,), (90, 86, 34), (90, 8
6, 85, 34, 59, 63, 93, 23), (90, 86, 85, 34, 36, 39, 56), (90, 86, 8
5, 34, 36, 1), (90, 86, 85, 34, 36, 59), (59,), (90, 86, 85, 34, 36,
59, 63, 23), (90, 86, 85, 34, 36), (90, 86, 85, 34, 39), (90, 86, 8
5, 34, 36, 59, 63, 93), (86, 34), (90, 86, 85, 34, 63), (86,), (90,
86, 85, 34, 36, 59, 93, 23, 39), (90, 86, 85), (90, 86, 85, 34, 59,
63), (90, 86, 85, 34), (34,), (90, 86, 85, 34, 36, 59, 93, 2), (90,
86, 85, 34, 36, 110), (90, 86), (90, 86, 85, 34, 59, 63, 93, 23, 2),
(90, 86, 85, 34, 36, 59, 93, 23, 63), (90, 86, 85, 34, 36, 59, 93, 2
3, 2), (90, 86, 85, 34, 59, 63, 93), (90, 34), (90, 86, 85, 34, 36,
59, 93, 23), (90, 85, 34), (90, 86, 85, 34, 36, 59, 63), (90, 85, 3
4, 36), (86, 85), (90, 86, 85, 34, 53), (85,), (90, 86, 85, 34, 59),
(90, 86, 85, 34, 36, 39), (90, 86, 85, 36), (86, 85, 34), (90, 86, 8
5, 34, 36, 59, 93), (90, 86, 85, 34, 36, 1, 24)}
==================================================
```

Sales performance evaluatiion

In [9]:

```
import json
import os

print(f"Sales performance of case {index_case} with retention rho =
unique_TOI = {item for t in set_TOI for item in t}
print(f"Unique items in TOI: {unique_TOI}")
print(f"Item rate = {len(unique_TOI)}/{len(var_unique_items)} = {le
total_TOI_volume = 0
total_TOI_revenue = 0.0
for item in unique_TOI:
    if (item,) in TDS:
        vals = TDS[(item,)]
        total_TOI_volume += vals[3]
        total_TOI_revenue += vals[5]
total_volume = sum(vals[3] for vals in TDS.values())
```

```
total_revenue = sum(vals[5] for vals in TDS.values())
print(f"Total sales volume of TOI : {total_TOI_volume}")
print(f"Sales volume rate of TOI  : {total_TOI_volume/total_volume:
print(f"Total sales revenue of TOI: {total_TOI_revenue:.4f}")
print(f"Sales revenue rate of TOI : {total_TOI_revenue/total_revenu

file_path = f"Sales_Performance_{dbName}_{index_case}_Retention_{rh
key = "C" + str(index_case)
value = [
    list(unique_TOI),                       # unique items in TOI
    len(unique_TOI)/len(var_unique_items),  # item rate
    total_TOI_volume,                       # total volume of TOI
    total_TOI_revenue,                     # total revenue of TOI
    total_TOI_volume/total_volume,          # volume rate
    total_TOI_revenue/total_revenue         # revenue rate
]

if os.path.exists(file_path):
    with open(file_path, "r") as f:
        data = json.load(f)
else:
    data = {}

data[key] = value
with open(file_path, "w") as f:
    json.dump(data, f, indent=4)
#####################################################################
```

```
Sales performance of case 9 with retention rho = 0.13:
Unique items in TOI: {1, 34, 2, 36, 39, 110, 85, 86, 23, 56, 53, 90,
59, 93, 24, 63}
Item rate = 16/119 = 0.1345
Total sales volume of TOI : 264151
Sales volume rate of TOI  : 0.4634
Total sales revenue of TOI: 688337.4392
Sales revenue rate of TOI : 0.3984
```

Radar chart of sales-oriented strategies

In [10]:

```
import numpy as np
import matplotlib.pyplot as plt
import os
import json

# ============================================================
# LOAD DATA
# ============================================================

if os.path.exists(f"Average_Bias_Weights_{dbName}.json"):

    with open(f"Average_Bias_Weights_{dbName}.json", "r") as f:
        data = json.load(f)
```

```
else:

    data = {}

# ============================================================
# LABELS
# ============================================================

labels = np.array([
    'Volume',
    'Occupancy',
    'Revenue'
])

# ============================================================
# STATS
# ============================================================

stats = {
    r'$\mathcal{C}_1$': data["C1"] if "C1" in data else [0, 0, 0],
    r'$\mathcal{C}_2$': data["C2"] if "C2" in data else [0, 0, 0],
    r'$\mathcal{C}_3$': data["C3"] if "C3" in data else [0, 0, 0],
    r'$\mathcal{C}_4$': data["C4"] if "C4" in data else [0, 0, 0],
    r'$\mathcal{C}_5$': data["C5"] if "C5" in data else [0, 0, 0],
    r'$\mathcal{C}_6$': data["C6"] if "C6" in data else [0, 0, 0],
    r'$\mathcal{C}_7$': data["C7"] if "C7" in data else [0, 0, 0],
    r'$\mathcal{C}_8$': data["C8"] if "C8" in data else [0, 0, 0],
    r'$\mathcal{C}_9$': data["C9"] if "C9" in data else [0, 0, 0],
    r'$\mathcal{C}_{10}$': data["C10"] if "C10" in data else [0, 0,
}

# ============================================================
# ANGLES
# ============================================================

num_vars = len(labels)

angles = np.linspace(
    0,
    2 * np.pi,
    num_vars,
    endpoint=False
).tolist()

angles += angles[:1]

# ============================================================
# FIGURE
# ============================================================

fig, ax = plt.subplots(
    figsize=(8, 8),
    subplot_kw=dict(polar=True)
)

# ============================================================
```

```
# STYLE
# ============================================================

colors = [
    '#1f77b4',
    '#ff7f0e',
    '#2ca02c',
    '#d62728',
    '#9467bd',
    '#8c564b',
    '#e377c2',
    '#7f7f7f',
    '#bcbd22',
    '#17becf'
]

markers = [
    'D',
    'x',
    '+',
    'o',
    's',
    'P',
    '*',
    'H',
    'v',
    '^'
]

lines = [
    'solid',
    'dashed',
    'dotted',
    'dashdot',
    (0, (3, 1, 1, 1)),
    (0, (5, 1)),
    (0, (1, 1)),
    (0, (3, 5, 1, 5)),
    (0, (5, 10)),
    (0, (1, 10))
]

# ============================================================
# PLOT
# ============================================================

for i, (name, avg_bias_weights) in enumerate(stats.items()):

    values = avg_bias_weights + avg_bias_weights[:1]

    ax.plot(
        angles,
        values,
        color=colors[i],
        linestyle=lines[i],
        linewidth=1.5,
```

```
        marker=markers[i],
        label=name
    )

    ax.fill(
        angles,
        values,
        color=colors[i],
        alpha=0.1
    )

# ============================================================
# RADAR ORIENTATION
# ============================================================

ax.set_theta_offset(np.pi / 2)
ax.set_theta_direction(-1)

# ============================================================
# AXIS LABELS
# ============================================================

ax.set_thetagrids(
    np.degrees(angles[:-1]),
    labels
)

# ============================================================
# TĂNG KÍCH THƯỚC LABEL:
# Volume / Occupancy / Revenue
# ============================================================

for label in ax.get_xticklabels():

    label.set_fontsize(20)
    label.set_fontweight('bold')

# ============================================================
# Y AXIS RANGE
# ============================================================

ax.set_ylim(
    0.8,
    0.94
)

# ============================================================
# TĂNG KÍCH THƯỚC CÁC SỐ LIỆU
# TRÊN CÁC VÒNG RADAR
# ============================================================

ax.tick_params(
    axis='y',
    labelsize=20
)
```

```
# ============================================================
# LEGEND
# ============================================================

legend_handles, legend_labels = ax.get_legend_handles_labels()

legend = fig.legend(
    legend_handles,
    legend_labels,

    loc="lower center",

    title=f"{dbName}",
    title_fontsize=22,

    ncol=5,

    columnspacing=0.5,   # <-- giảm khoảng cách giữa các cột
    handletextpad=0.4,   # khoảng cách giữa ký hiệu và text

    frameon=True,
    facecolor="white",
    edgecolor="black",
    framealpha=1.0,

    bbox_to_anchor=(0.5, 0.08),

    fontsize=24
)
legend.get_title().set_fontweight("bold")


# ============================================================
# LAYOUT
# ============================================================

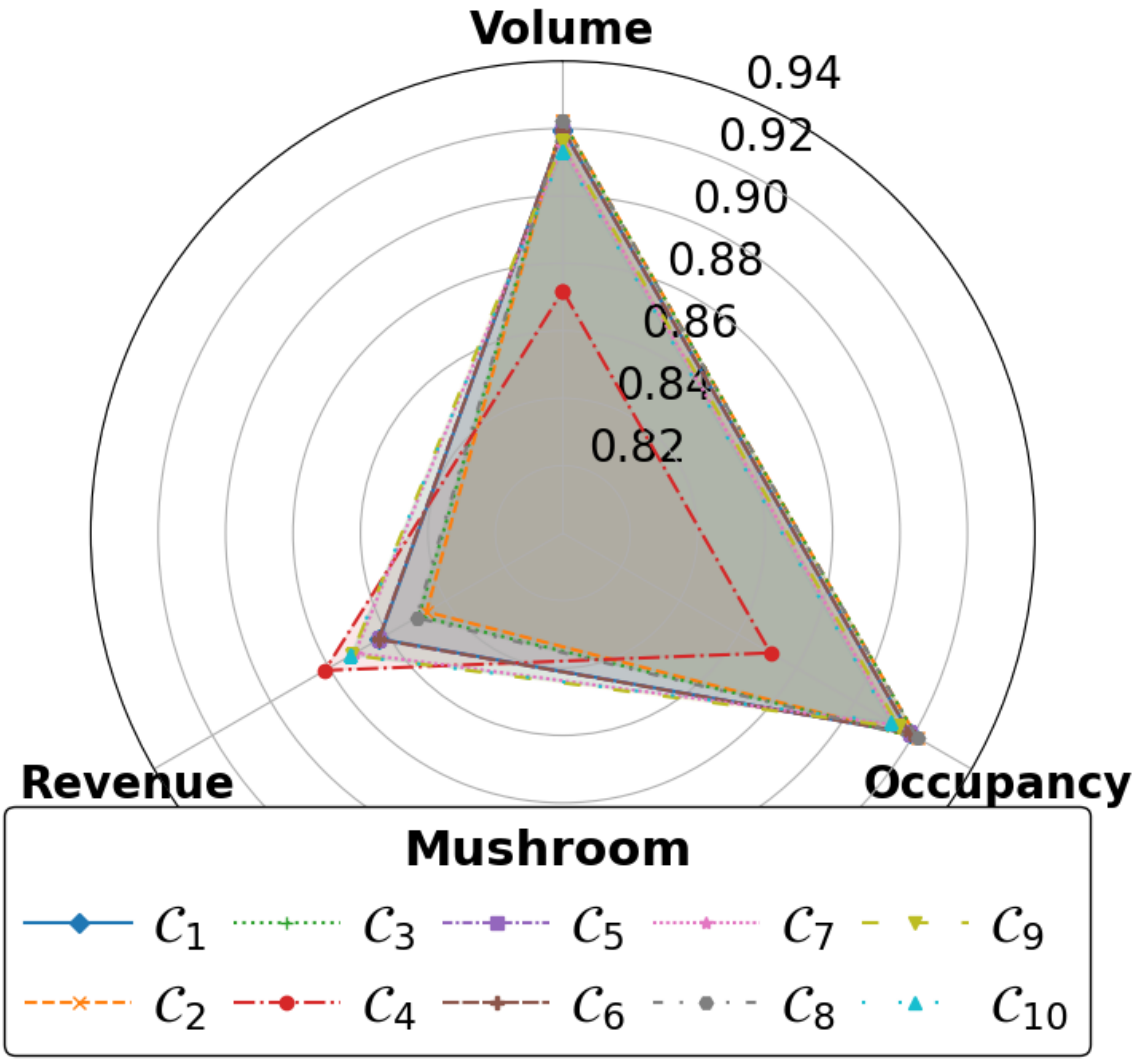
plt.subplots_adjust(
    bottom=0.14
)

# ============================================================
# SAVE
# ============================================================

plt.savefig(
    f"Radar_Chart_{dbName}.pdf",
    dpi=300,
    bbox_inches="tight"
)

# ============================================================
# SHOW
# ============================================================

plt.show()
```



Entropy-based weights

In [11]:
```python
import numpy as np


def entropy_based_thresholds(X):
    """
    Compute entropy-based thresholds for multi-criteria evaluation.

    Parameters
    ----------
    X : array-like of shape (m, k)
        Normalized decision matrix.
        m: number of bias strategies
        k: number of criteria (e.g., Support, Occupancy, Profit)

    Returns
    -------
    weights : numpy.ndarray of shape (k,)
        Entropy-based weights (thresholds) for each criterion.
    """
    X = np.array(X, dtype=float)
```

```
    m, k = X.shape

    # Step 1: probability distribution
    column_sums = X.sum(axis=0)
    P = X / column_sums

    # Step 2: entropy computation
    omega = 1.0 / np.log(m)
    entropy = -omega * np.sum(P * np.log(P), axis=0)

    # Step 3: diversification degree
    d = 1.0 - entropy

    # Step 4: entropy weights (thresholds)
    weights = d / np.sum(d)

    return weights


X = np.array(list(data.values()))
alpha, beta, gamma = entropy_based_thresholds(X)

print("Entropy-based weights (alpha, beta, gamma):")
print("alpha :", alpha)
print("beta  :", beta)
print("gamma :", gamma)
```

```
Entropy-based weights (alpha, beta, gamma):
alpha : 0.3796218337534444
beta  : 0.3771202988255723
gamma : 0.24325786742098335
```

The optimal sales-oriented strategy selection

In [12]:

```
import numpy as np

# 1. Data matrix (7 strategies x 3 criteria)
data = np.array(list(data.values()))

# POSITIVE CRITERIA (Benefit criteria)
is_benefit = [True, True, True]

def calculate_entropy_weights(matrix):
    # Normalize the decision matrix to get probabilities
    norm_matrix = matrix / matrix.sum(axis=0)

    # Calculate Entropy (e)
    k = 1.0 / np.log(len(matrix))
    # Avoid log(0) by adding a small constant (1e-12)
    entropy = -k * np.sum(norm_matrix * np.log(norm_matrix + 1e-12)
    # Calculate the degree of diversification (d)
    dispersion = 1 - entropy
    # Calculate weights (w)
```

```
    weights = dispersion / np.sum(dispersion)
    return weights

def calculate_topsis(matrix, weights, is_benefit):
    # Vector Normalization
    norm_matrix = matrix / np.sqrt(np.sum(matrix**2, axis=0))

    # Multiply by weights to get the weighted normalized decision m
    weighted_matrix = norm_matrix * weights

    # Define ideal best (PIS) and ideal worst (NIS)
    pis = []
    nis = []
    for j in range(len(is_benefit)):
        if is_benefit[j]:
            pis.append(np.max(weighted_matrix[:, j]))
            nis.append(np.min(weighted_matrix[:, j]))
        else:
            pis.append(np.min(weighted_matrix[:, j]))
            nis.append(np.max(weighted_matrix[:, j]))

    # Calculate distances to PIS and NIS (Euclidean distance)
    dp = np.sqrt(np.sum((weighted_matrix - pis)**2, axis=1))
    dn = np.sqrt(np.sum((weighted_matrix - nis)**2, axis=1))

    # Calculate Closeness Coefficient
    score = dn / (dp + dn)
    return score


weights = calculate_entropy_weights(data)
scores = calculate_topsis(data, weights, is_benefit)

# Display results
print("--- TOPSIS score ---")
print(f"Entropy-based weights: {weights}")
print("-" * 30)
for i, s in enumerate(scores):
    print(f"Sales-oriented strategy C{i+1}: TOPSIS = {s:.8f}")

# Find the best configuration
best_idx = np.argmax(scores)
print("-" * 30)
print(f"The optimal sales-oriented strategy: C{best_idx + 1}")
```

```
--- TOPSIS score ---
Entropy-based weights: [0.37962183 0.37712029 0.24325788]
------------------------------
Sales-oriented strategy C1: TOPSIS = 0.83767820
Sales-oriented strategy C2: TOPSIS = 0.75053115
Sales-oriented strategy C3: TOPSIS = 0.76427916
Sales-oriented strategy C4: TOPSIS = 0.24940315
Sales-oriented strategy C5: TOPSIS = 0.83934420
Sales-oriented strategy C6: TOPSIS = 0.83934420
Sales-oriented strategy C7: TOPSIS = 0.83128456
Sales-oriented strategy C8: TOPSIS = 0.76955291
Sales-oriented strategy C9: TOPSIS = 0.85758661
Sales-oriented strategy C10: TOPSIS = 0.81219782
------------------------------
The optimal sales-oriented strategy: C9
```

In [13]:

```python
import networkx as nx
import matplotlib.pyplot as plt

nodes = TOI_nodes  # list các tuple

# ============================================================
# CONFIG
# ============================================================

# Dịch toàn bộ graph sang phải
X_SHIFT = 0.3

# Khoảng cách dọc giữa các level
y_gap = 0.18

# Khoảng cách ngang giữa tâm các node
# 0.002 là quá nhỏ -> gây chồng node
x_gap = 0.16

# Kích thước node
NODE_SIZE = 3000

# ============================================================
# PARENT MAP
# ============================================================

parent_map = {}

for child in nodes:

    if len(child) > 1 and child[:-1] in nodes:
        parent_map[child] = child[:-1]

    else:
        parent_map[child] = None

# ============================================================
# GRAPH
# ============================================================
```

```
G = nx.DiGraph()

# Root
G.add_node(dbName)

# Add nodes
for node in nodes:
    G.add_node(node)

# ============================================================
# LABELS
# ============================================================

labels = {}

for node in G.nodes():

    if node == dbName:

        labels[node] = dbName

    else:

        # Level 1
        if len(node) == 1:

            labels[node] = rf"$\langle${node[-1]}$\rangle$"

        # Level > 1
        else:

            labels[node] = rf"$\langle${node[-1]}$\rangle$"

# ============================================================
# EDGES
# ============================================================

for child, parent in parent_map.items():

    if parent:

        G.add_edge(parent, child)

    else:

        G.add_edge(dbName, child)

# ============================================================
# LEVELS
# ============================================================

levels = {
    0: [dbName]
}

for node in nodes:
```

```
    lvl = len(node)

    levels.setdefault(lvl, []).append(node)

# ============================================================
# NODE POSITION
# ============================================================

pos = {}

for lvl, level_nodes in levels.items():
    n = len(level_nodes)

    # Các node luôn được trải đều từ trái sang phải
    x_positions = [i * x_gap for i in range(n)]

    for i, node in enumerate(level_nodes):
        x = X_SHIFT + x_positions[i]
        y = -lvl * y_gap
        pos[node] = (x, y)

    # ------------------------------------------------------
    # Gán vị trí
    # ------------------------------------------------------

    for i, node in enumerate(level_nodes):

        x = x_positions[i] + X_SHIFT

        y = -lvl * y_gap

        pos[node] = (x, y)

# ============================================================
# DRAW
# ============================================================

fig, ax = plt.subplots(figsize=(7, 11.5))

nx.draw(
    G,
    pos,
    labels=labels,

    # ------------------------------------------------------
    # Node
    # green ----------------------------------------------------

    node_size=NODE_SIZE,
    node_color="none",
    edgecolors="black",
    # ------------------------------------------------------
    # Font
    # ------------------------------------------------------
```

```
    font_size=20,
    font_weight="bold",

    # ------------------------------------------------------
    # Edge
    # ------------------------------------------------------

    arrowsize=10,

    ax=ax
)

# ============================================================
# CASE
# ============================================================

fig.text(
    0.5,
    0.945,

    rf"Case $\mathcal{{C}}_{{{index_case}}}$",

    ha="center",
    va="top",

    fontsize=20,
    fontweight="bold"
)

# ============================================================
# RANKING
# ============================================================

fig.text(
    0.18,
    0.92,

    rf"$1st \qquad\quad "
    rf"2nd \qquad\quad "
    rf"3rd \qquad\quad "
    rf"4th \qquad\quad "
    rf"5th$",

    ha="left",
    va="top",

    fontsize=16,
    fontweight="bold"
)

# ============================================================
# LEVEL LABEL
# ============================================================
first_pos = 0.795
move_pos = 0.08955
for i in range(1, max_level + 1):
```

```
    fig.text(
        0.0,
        first_pos - (i - 1) * move_pos,

        rf"$\ell = {i}$",

        ha="left",
        va="top",

        fontsize=20,
        fontweight="bold"
    )



# ============================================================
# AXIS
# ============================================================

ax.margins(0)

ax.axis("off")

# ============================================================
# AXIS LIMITS
# ============================================================

xs = [x for x, y in pos.values()]
ys = [y for x, y in pos.values()]

# ------------------------------------------------------------
# Tính giới hạn dựa trên node thực tế
# ------------------------------------------------------------

x_min = min(xs)
x_max = max(xs)

y_min = min(ys)
y_max = max(ys)

# Khoảng đệm
x_margin = 0.18
y_margin = 0.08

ax.set_xlim(
    x_min - x_margin,
    x_max + x_margin
)

ax.set_ylim(
    y_min - y_margin,
    y_max + y_margin
)

# ============================================================
# LAYOUT
```

```
# ============================================================


plt.tight_layout(
    rect=[0, 0, 1, 0.94]
)

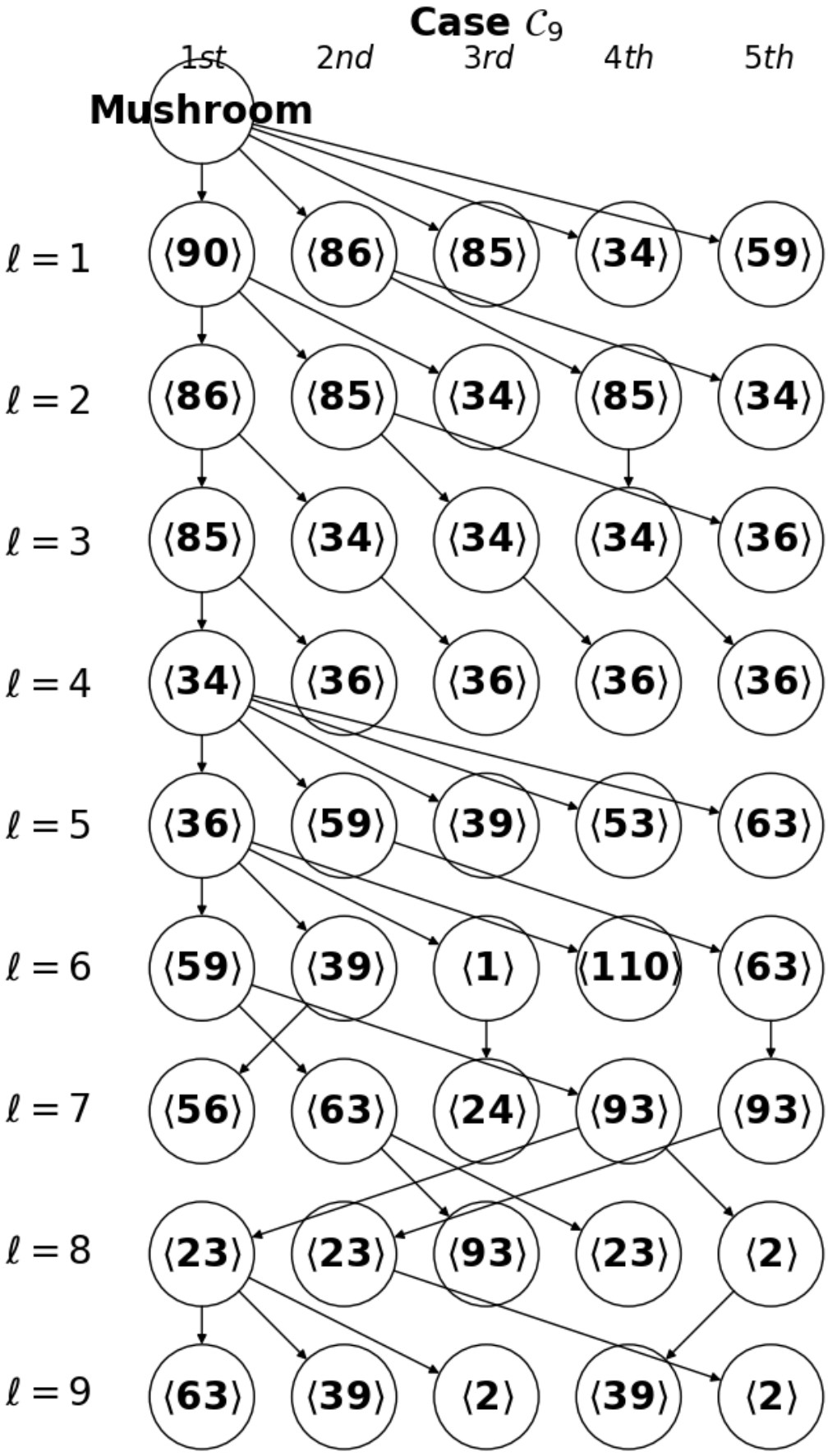
# ============================================================
# SAVE
# ============================================================

plt.savefig(
    f"TOI_Hierarchy_{dbName}_{index_case}.pdf",

    dpi=300,

    bbox_inches="tight"
)

plt.show()
```



In [14]:
```
# import kagglehub

# # Download latest version
# path = kagglehub.dataset_download("roopacalistus/superstore")
```

```
# print("Path to dataset files:", path)
```